Lecture 4 (Thu, May 21)

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Representations of Rotations

A rotation in three-dimensional space has three degrees of freedom. Several common ways to represent rotations are:

1. **Euler angles**, such as the ZYZ convention.
2. **Roll-pitch-yaw (RPY) angles**.
3. **Axis-angle representation**, using a unit axis

$$k = (k_x, k_y, k_z), \qquad \|k\| = 1,$$

   together with a rotation angle θ.

4. **Unit quaternions**, represented by four numbers:

$$Q = \left(\cos\frac{\theta}{2}, k_x \sin\frac{\theta}{2}, k_y \sin\frac{\theta}{2}, k_z \sin\frac{\theta}{2}\right).$$

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[2] Euler angles and RPY angles are minimal because they use three parameters, but they can have singularities. Unit quaternions are not minimal, since they use four numbers with a unit-norm constraint.

A homogeneous transformation matrix $H_n^0(q)$ represents the pose of frame n relative to frame 0. It contains both a rotation and a translation:

$$H_n^0(q) \in SE(3) \subset \mathbb{R}^{4\times 4}.$$

The group $SE(3)$, called the special Euclidean group, is the set of rigid body motions in three dimensions.

For example, if $\bar{p}^2$ is a point written in homogeneous coordinates in frame 2, then

$$H_1^0 H_2^1 \bar{p}^2 = H_1^0 \bar{p}^1 = \bar{p}^0 = H_2^0 \bar{p}^2,$$

where

$$H_2^0 = H_1^0 H_2^1.$$

The **direct kinematics map** is

$$H_n^0 : q \mapsto H_n^0(q), \qquad \mathbb{R}^n \to SE(3).$$

For a serial manipulator,

$$H_n^0(q) = H_1^0(q_1) H_2^1(q_2) \cdots H_n^{n-1}(q_n).$$

Denavit–Hartenberg Parameters

The Denavit–Hartenberg convention gives a systematic way to assign coordinate frames to links and joints.

For the link between joint $i-1$ and joint i, define a reference frame

$$o_{i-1}x_{i-1}y_{i-1}z_{i-1}.$$

The z_{i-1}-axis is chosen along the axis of joint i, while the x-axis is chosen along the common normal between consecutive joint axes.

./lecture-4img/05-21-joint.png

Figure 1: Reference frame assignment for consecutive joints under the Denavit–Hartenberg convention.

To move from frame $i-1$ to frame i, we use four transformations:

1. Translate by d_i along z_{i-1}.
2. Rotate by θ_i about z_{i-1}.
3. Translate by a_i along x_i.
4. Rotate by α_i about x_i.

The four DH parameters are:

- d_i: offset along z_{i-1}.
- θ_i: rotation angle about z_{i-1}.
- a_i: link length along the common normal.
- α_i: twist angle about x_i, aligning z_{i-1} with z_i.

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Using the standard DH convention, the homogeneous transformation is

$$H_i^{i-1}(q_i) = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i).$$

In matrix form,

$$H_i^{i-1}(q_i) = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Multiplying these matrices gives

$$H_i^{i-1}(q_i) = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

[3] For a revolute joint, θ_i is the joint variable. For a prismatic joint, d_i is the joint variable. The remaining DH parameters are fixed by the robot geometry.

Non-Uniqueness of the DH Convention

The DH convention is useful, but the frame assignment is not always unique. Non-uniqueness can occur in the following cases:

1. The base frame, since there is no previous link.
2. The end-effector frame, since there is no next link.
3. Parallel consecutive joint axes.
4. Intersecting consecutive joint axes.
5. Prismatic joints.

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For prismatic joints, the displacement d_i changes with the joint variable, rather than the angle θ_i. Therefore, extra care is needed when identifying which DH parameter is variable.

[4] If consecutive joint axes are parallel, there are infinitely many possible common normals. If they intersect, the common normal has zero length, so the frame assignment can become ambiguous.